



Tuples

Practical Skills Module 2: Topic 7

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Topic Code	PM-02-PS07
Topic	Use Python tuple to store multiple items in a single variable

Scope of Practical Skill

Given an applicable instruction and access to a learning platform, the learner must be able to:

- PA0701 Pack and unpack tuples
- PA0702 Compare tuples
- PA0703 Use tuples as keys in dictionaries
- PA0704 Delete tuples
- PA0705 Slice tuples

Applied Knowledge

- AK0701 Concept, definition and functions of Python data types

Internal Assessment Criteria

- IAC0701 Expected results with Python tuple are achieved

Resources:

Knowledge	Information from Python Programmer Curriculum
National Curriculum Framework	900221-000-00-PM-02



Practical task

Follow the facilitator's instructions to complete the following activities:

Scenario: E-Commerce Inventory Management System

In this scenario, you are tasked with developing a Python program to manage product data for an e-commerce platform. Each product's information, such as its name, category, and stock, is stored as tuples. The program will allow you to perform operations like packing and unpacking tuples, comparing them to sort inventory, using tuples as keys in a dictionary for quick lookups, deleting tuples for discontinued products, and slicing tuples to extract specific data.

Create a Python file called **product_inventory.py** and complete the steps below:

Step 1: Pack and unpack tuples.

- Create a tuple to represent a product using the following details:
 - Product name (string)
 - Category (string)
 - Stock quantity (integer)

Example structure: ("Bluetooth Speaker", "Electronics", 120)

- Unpack the tuple into separate variables and print each variable with a label.
 - *Expected output:*

```
Product Name: Bluetooth Speaker
Category: Electronics
Stock: 120
```

Step 2: Compare tuples to sort by stock quantity.

- Create a list containing at least **three product tuples** with the same structure.
- Sort the list of tuples **by the stock quantity** (which is the third item in the tuple).
- Print the sorted list to show the products ordered from lowest to highest stock.

Tip: You may use a sorting method that sorts based on the third index of each tuple.

Step 3: Use tuples as keys in a dictionary.

- Use each product's **(name, category)** as the key in a dictionary.
- Store the stock quantity as the value for each key.
- Print the dictionary to show product details mapped to their current stock.

Tip: This simulates quick lookup by product identity.

Step 4: Delete tuples from list or dictionary.

- Choose one product that is now **discontinued**.
- Remove the associated tuple from:
 - The original **list of tuples**
 - The **dictionary** created in Step 3
- Print both updated structures to confirm that the product has been removed.

Step 5: Slice tuples to extract specific details.

- From any product tuple, use **tuple slicing** to extract only the **product name and category**, ignoring the stock quantity.
- Print the sliced result to verify that only the first two elements are included.
 - *Expected output:* A tuple like ("Bluetooth Speaker", "Electronics")



Take note

Important: Be sure to upload all files required for the task submission inside your task folder and then click "Request review" on your dashboard.
